s6_beta_overlay

equity strategy | family: equity-overlay | status: validated | derived from `s6_neutral_levered`

What this strategy does

Strategy 6 + synthetic market position of 0.5x trailing NAV

Why it should work. Alpha book earns the drift; the overlay earns the equity risk premium on the same capital.

Design notes (from the strategy's source file).

Strategy 6 + 0.5x market beta overlay.

Keeps Strategy 6's alpha book completely untouched and holds a separate synthetic market-index position of 0.5x trailing NAV beside it (re-sized quarterly, 2bps cost) -- the 130/30-style long-extension structure real funds use to add market exposure without distorting the alpha signal. This is how the equity side clears the S&P 500's return in this sample.

parameter	value
base_unit_fraction	0.003
max_gross_leverage	2.5
liquidity_cap_frac	0.05
return_basis	mktadj
cap_mode	priority
beta_overlay	0.5

Performance

metric	s6 beta overlay	market (CRSP VW)
Annualized return	10.08%	9.94%
Annualized volatility	10.51%	19.35%
Sharpe (NAV-pct)	0.97	0.59
Sortino	1.40	
Max drawdown	-23.2%	-55.5%
Calmar	0.43	
Final NAV ($10M start)	$64,912,423	
Worst day	-4.66%	
% positive days	55.4%	
Sharpe 1996-2001	1.30	
Sharpe 2002-2007	1.14	
Sharpe 2008-2013	0.48	

Correlation of daily returns with the market: **0.94**.

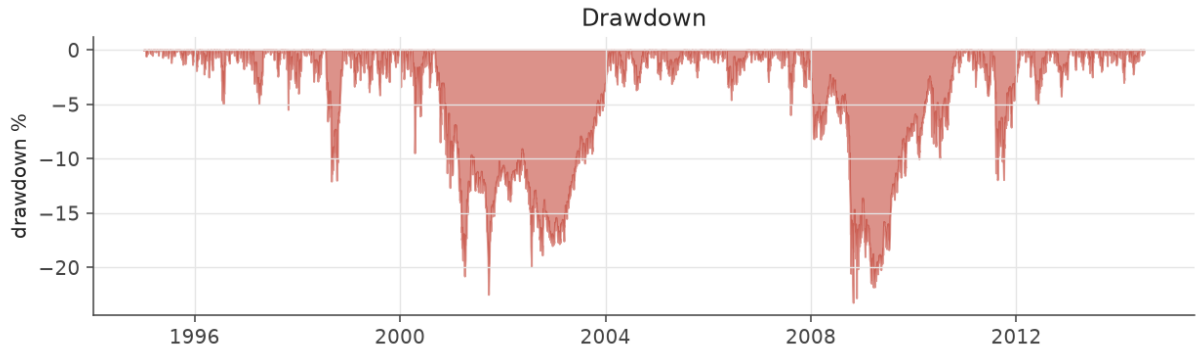
• best Sharpe among equity: `s10_vol_rank_tilt` (1.81); this one: 0.966

• best return among equity: `s6_beta_overlay_aggressive` (0.15); this one: 0.101

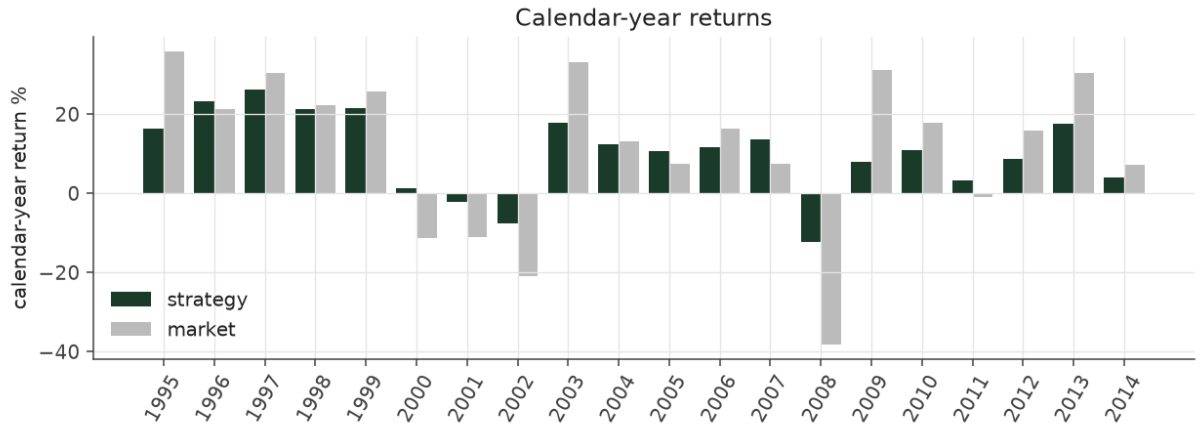
• best Calmar among equity: `s10_vol_rank_tilt` (0.614); this one: 0.434



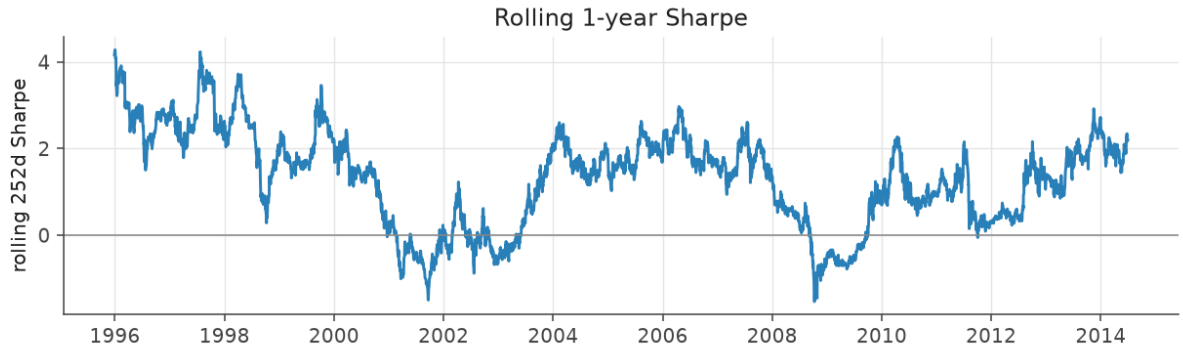
NAV (log scale)



Drawdown from running peak

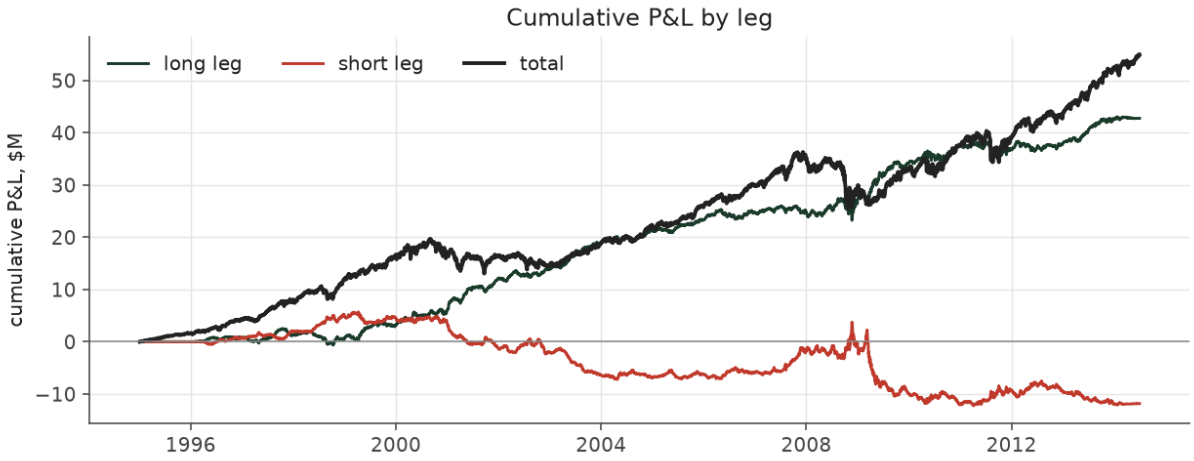


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	42.74	-11.85
P&L after costs ($M)	39.45	-15.40
of which common (average-event) component ($M)	26.88	-27.72
of which excess / PEAD component ($M)	15.86	15.87
excess minus costs ($M)	12.57	12.32
gross return per $ traded (bp)	253.36	-68.44
positions in the money (hit rate)	0.56	0.49
stand-alone Sharpe (leg only)	1.20	-0.24

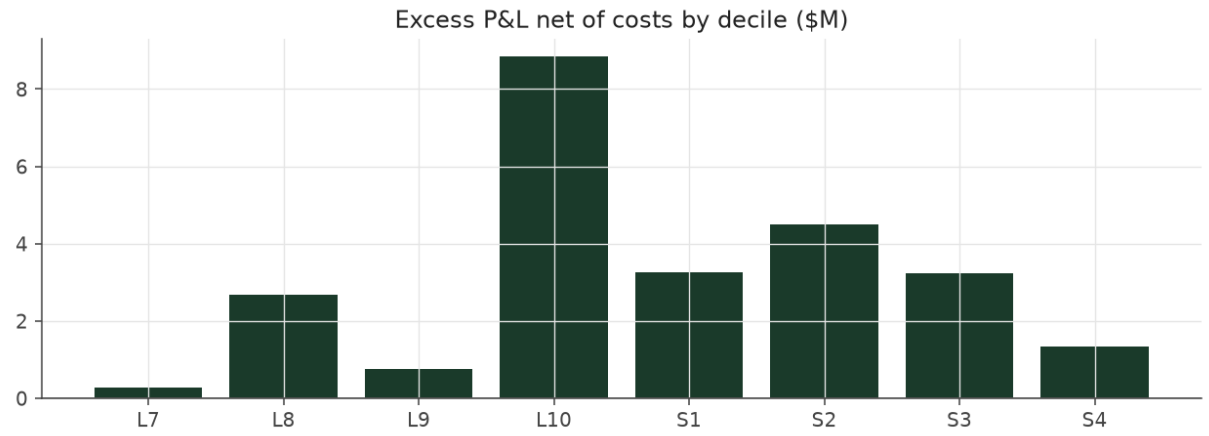
Daily P&L correlation between the two legs: **-0.81**.

• **The short leg loses money in absolute terms** (-11.8M before costs) while the long leg earns +42.7M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 26.9M / 27.7M. Stripping that common component, the surprise signal itself earned +15.9M on the long side and +15.9M on the short side.

• The two legs' daily P&L are -0.81 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 1.20 long / -0.24 short vs 0.97 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	7	8,029	148.61	3.03	0.56	0.29	0.27	1.1%	18.00
long	8	16,588	394.76	9.19	3.47	0.78	2.69	10.8%	68.11
long	9	17,747	522.21	8.37	1.77	1.01	0.76	3.1%	14.60
long	10	17,174	621.45	18.85	10.07	1.22	8.85	35.6%	142.39
short	1	17,169	624.20	-6.03	4.58	1.33	3.25	13.1%	52.06
short	2	17,281	539.87	-4.65	5.58	1.09	4.49	18.0%	83.16
short	3	15,970	404.59	-3.39	4.05	0.81	3.24	13.0%	79.97
short	4	9,171	162.59	-1.33	1.66	0.31	1.35	5.4%	82.78



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	14,213	306.96	5.45	7.41	1.18	6.24	25.1%	203.15
2	22,885	641.27	6.15	8.35	1.80	6.55	26.3%	102.10
3	25,501	785.23	6.22	7.75	1.64	6.11	24.6%	77.86
4	27,583	839.10	3.71	4.93	1.31	3.62	14.5%	43.15
5	28,947	845.72	2.51	3.30	0.93	2.37	9.5%	28.03

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	13,650	225.40	5.20	6.74	1.13	5.61	22.6%	249.07
2	23,084	676.60	5.79	8.72	2.03	6.69	26.9%	98.85
3	25,851	834.81	6.30	8.50	1.67	6.83	27.5%	81.85
4	27,665	843.84	4.54	5.11	1.18	3.93	15.8%	46.53

5	28,874	837.44	2.21	2.66	0.84	1.82	7.3%	21.71

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	45,855	1,296.80	-0.36	7.02	2.28	4.74	19.1%	36.57
41-60d	25,652	773.42	4.67	4.86	1.58	3.28	13.2%	42.46
61-90d	35,324	1,006.66	7.13	8.63	2.12	6.51	26.2%	64.70
>90d	12,298	341.40	12.61	11.22	0.87	10.35	41.6%	303.13

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	33,895	702.89	9.20	10.91	1.38	9.54	38.3%	135.71
2002-07	39,956	1,131.98	10.13	12.35	2.22	10.13	40.7%	89.47
2008-13	45,278	1,583.40	4.72	8.47	3.25	5.22	21.0%	32.97

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	22,094	335.42	1.77	2.48	0.71	1.77	7.1%	52.89
Chems	2,590	227.57	1.44	1.85	0.40	1.45	5.8%	63.75
Durbl	2,843	245.81	2.39	2.91	0.45	2.46	9.9%	100.13
Enrgy	6,194	278.68	3.95	4.57	0.52	4.05	16.3%	145.38
Hlth	11,696	325.36	2.67	3.38	0.68	2.69	10.8%	82.79
Manuf	12,034	304.84	1.76	2.55	0.63	1.92	7.7%	62.91
Money	17,583	299.07	2.59	3.36	0.67	2.69	10.8%	89.95
NoDur	5,081	274.80	1.38	1.87	0.52	1.35	5.4%	49.05
Other	18,377	325.08	2.31	3.13	0.72	2.41	9.7%	74.15
Shops	11,984	317.62	3.13	3.82	0.65	3.17	12.7%	99.77
Telcm	3,981	270.57	0.18	0.75	0.49	0.26	1.0%	9.46
Utils	4,672	213.45	0.48	1.06	0.39	0.67	2.7%	31.19

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	39,523	1,094.31	4.11	7.37	1.98	5.39	21.7%	49.25
low vol	40,091	1,165.68	10.19	12.82	2.56	10.26	41.2%	87.97
mid vol	37,903	1,105.61	9.59	11.28	2.19	9.09	36.5%	82.20

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	10.08%	0.00%
0.5%	1.8	9.93%	0.15%
1.0%	3.5	9.77%	0.31%
2.0%	7.0	9.44%	0.64%
5.0%	17.5	8.32%	1.76%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.